Dwight Look College of
ENGINEERING
TEXAS A&M UNIVERSITY

Sound Localization Robot, Team 5

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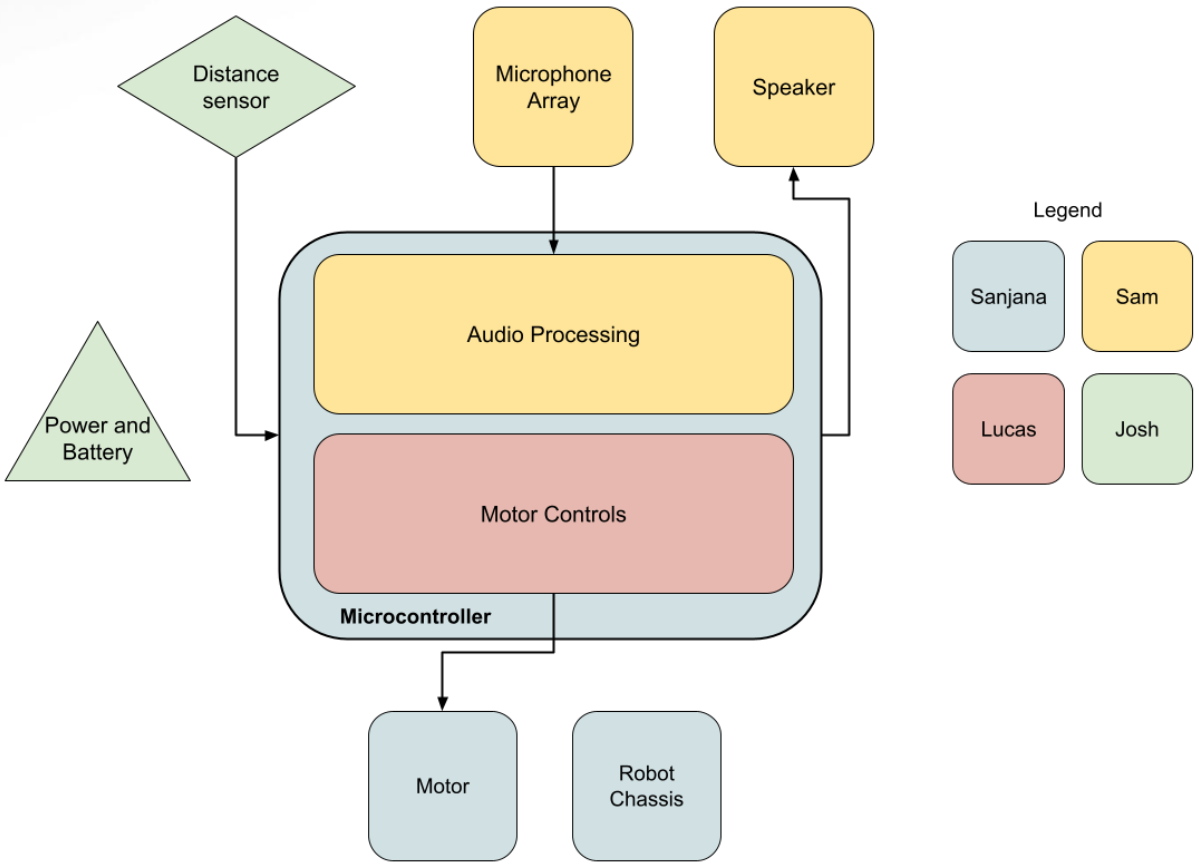
- **Problem statement:** First responders will often end up in situations where their vision and ability to navigate the scene become heavily obstructed

Figure 1. First responder crawls through smoke and broken glass to rescue elderly man from house fire (Independent TV)

Solution proposal:

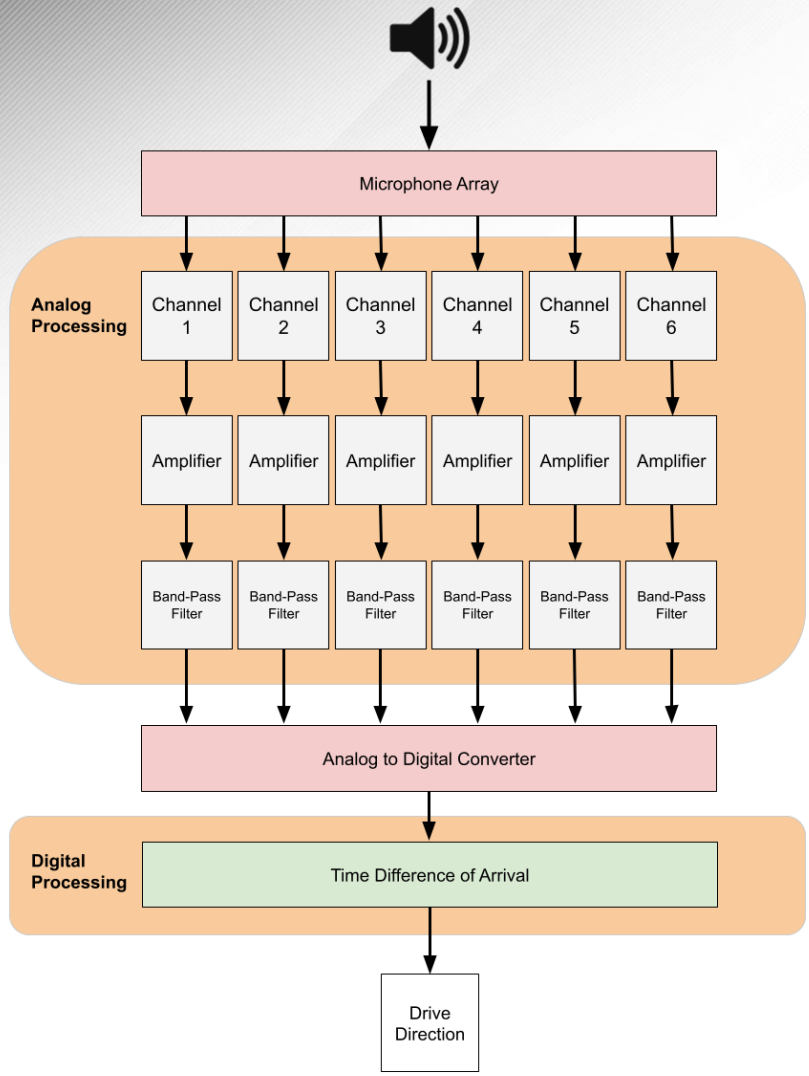
- Develop a small robot that is can move in all four directions
- Locate and Maneuver to sound source
- Ignore background and machine noise
- Alert when reaches sound source

Robot Sub Systems:

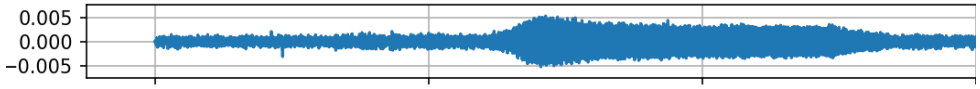


Input Signal Software

Samuel Lackey

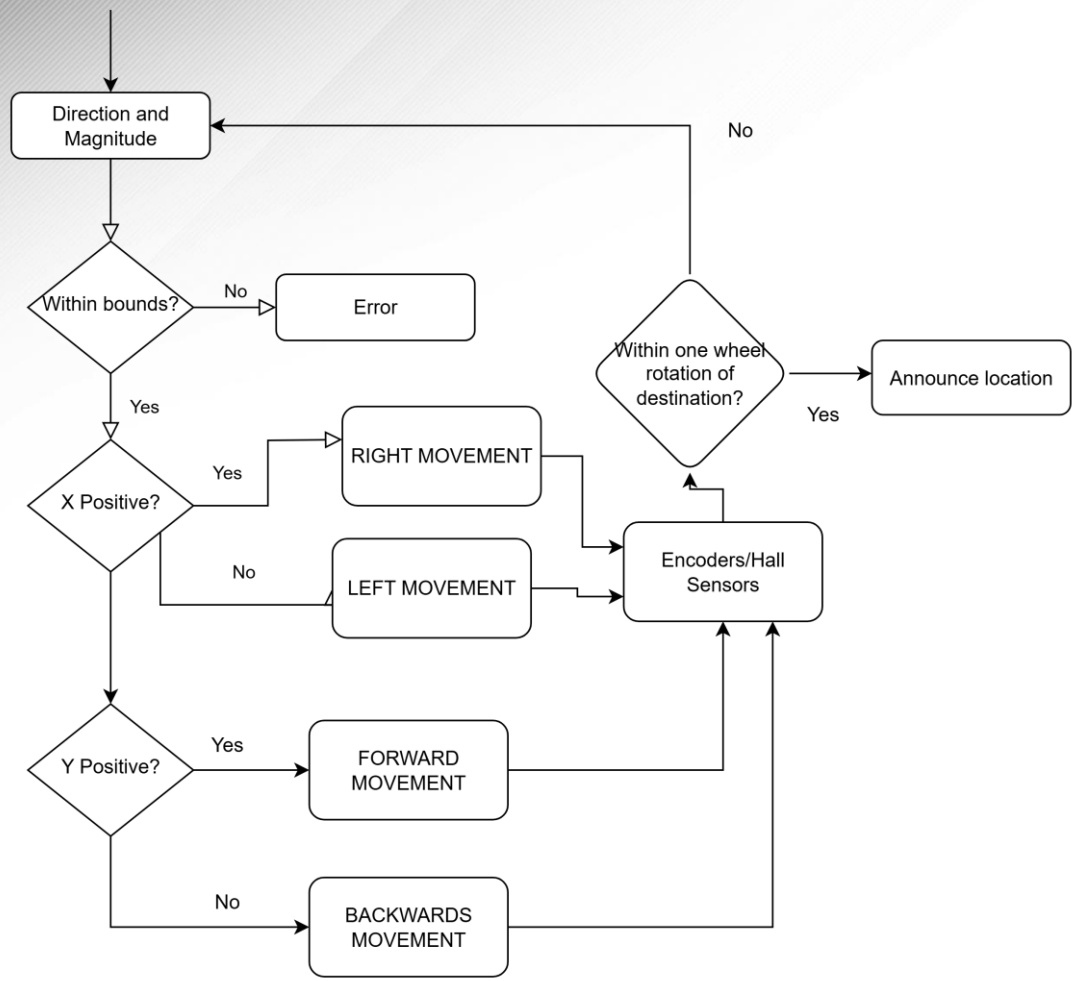


Accomplishments since start: 20 Hours
• Functioning test of microphone • Research of TDOA Algorithms
Ongoing progress/problems and plans until the next presentation:
• Lots of background noise • Mics are close, limits TDOA algorithm



Navigation Software

Lucas Kettler



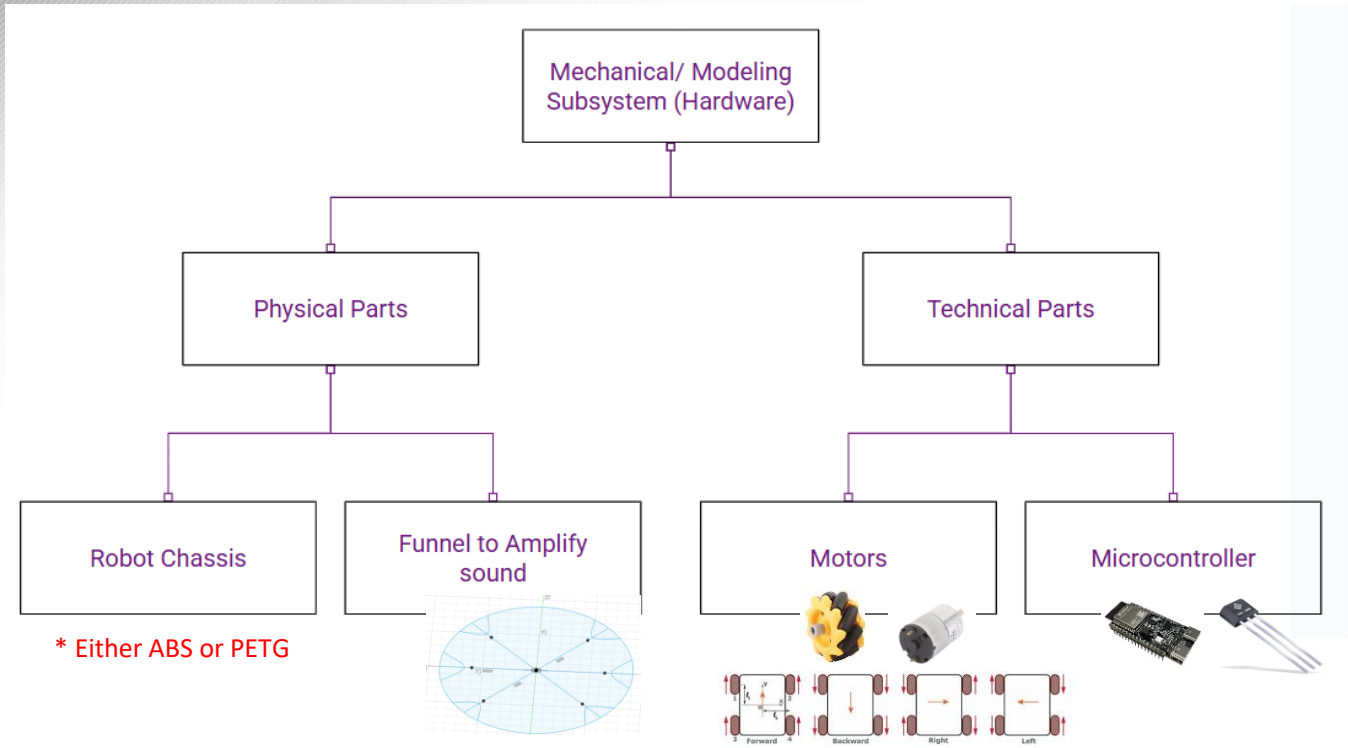
Accomplishments since start:
18 Hours

- Full length pseudocode for simple pathing

Ongoing progress/problems and plans until the next presentation:

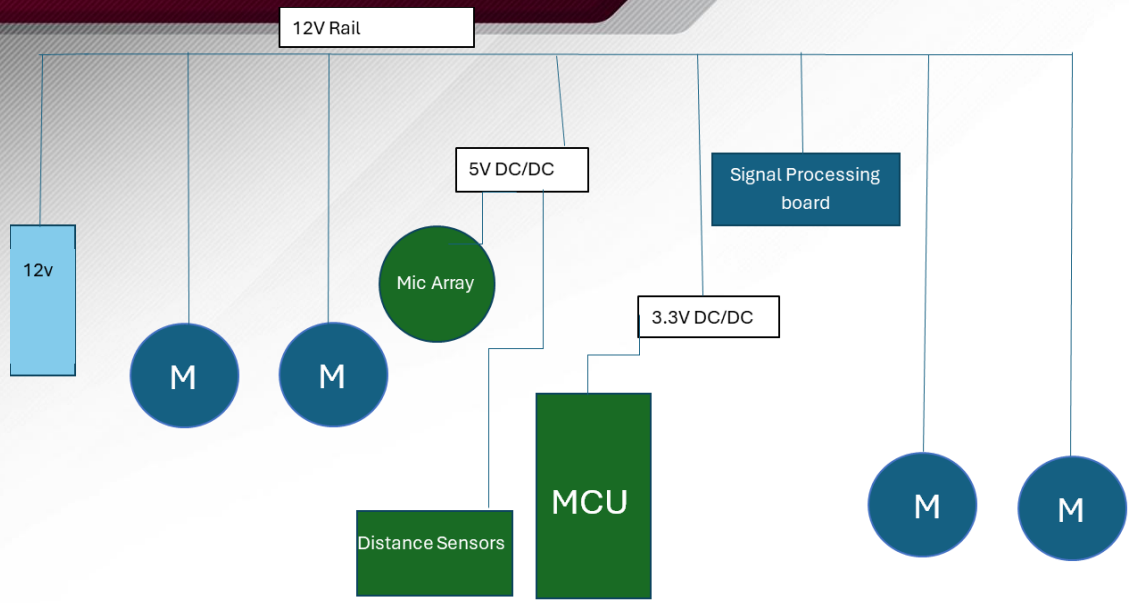
- Finished code ready for testing/debugging
- Pseudocode for obstacle avoidance

Mechanical/ Modeling Subsystem



- Take in directions from the navigation system, and process it into instructions that the microcontroller can do
- Takes in instructions from microcontroller to control the motors and steer wheels in direction of sound
- Build funnel to amplify the sound that reaches the microphones

Power System and Distance Sensors



Power System

Joshua Teele

- Subsystem powers every electrical component
- Use of DC/DC buck converters to power low-power electronics
- Switch near the source to turn robot on and off

Chassis

Ultrasonic Sensor

Distance Sensors System

- System will be on every side of the chassis
- Will be used to make sure robot does not run into anything

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Execution plan

Work	DUE DATE	09/08/25	9/15/25	09/22/25	9/29/25	10/06/25	10/13/25	10/20/25	10/27/25	11/03/25	11/10/25
ConOps	9/17/2025	Team					Not Started				
Functioning test of Microphone	9/20/2025		Sam				Behind Schedule				
3D model first design of Mic Funnel			Sanjana				Completed				
Functioning test of a motor system			Lucas	Lucas	Lucas		In Progress				
Firmware Logic/Pseudocode			Lucas								
Preliminary power management block diagram			Josh								
Midterm Presentation	9/22/2025		Team								
3D Printing of Microphone Funnel	9/28/2025		Sanjana								
Logic for TDOA code					Sam						
Preliminary Bill of Materials	9/26/2025		Team								
Order materials & equipment	9/26/2025				Team						
Preliminary power budget	09/22/2025			Josh							
Bandpass filter	10/06/2025			Sam							
FSR	10/04/2025			Team							
ICD	10/04/2025			Team							
TDOA Algrorithm	10/04/2025			Sam							
Milestones	10/04/2025			Team							
Validation Plan	10/04/2025			Team							
Status Update Presentation	10/20/2025						Team				
Final Audio Processing	11/1/2025						Sam				
Final Motor Drive	11/1/2025						Lucas				
Final Power Test	11/1/2025						Josh				
Final Motor Control Signal Verification	11/1/2025						Sanjana				
Final Presentation	11/10/2025									Team	

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Validation plan

Test Name	Success Criteria	Methodology	Status	Engineer(s) Responsible
Microphone Operating	Read analog audio from UMA-8 Mic Array	A 440Hz sound is played from differing locations and distances from the mic. Volume and some direction will be seen using a Python Script	Success	Sam Lackey
Funnel Operation	Amplify sound reaching microphone	Use software to see sound arcs reaching microphone, then 3D-print to test on actual microphone array	Untested	Sanjana Pasumarthi
Test Motor Direction	Wheels can move in all 4 directions	Take in instructions from microcontroller, and make sure the circuit works	Untested	Sanjana Pasumarthi
Battery Charge	Battery charges and decharges	Charge and uncharge while measuring current and voltage compared to spec sheet	Untested	Joshua Teele
Power System Test	With the system under a simulated full load, all regulated voltage outputs (e.g., 3.3V, 5V, 12V) must remain stable and within **±5%** of their target values.	Using a multimeter and power resistors to simulate load, measure the voltage at each regulator's output on the PCB. Verify stability and accuracy against the design specifications.	Untested	Joshua Teele
Motor Control	Motor spins at a set RPM	Motor is connected to a simple controller and powered on	Untested	Lucas Kettler
Distance Sensor	Sensor detects a wall ~6 inches away	Sensor is connected to a simple independent circuit, and brought to a measured distance from a wall	Untested	Joshua Teele
Final Audio Processing	Sound angle can be detected within 3 degrees	Play a 440Hz sound at different predetermined angles around the microphone array, compared final output to tested angle	Untested	Samuel Lackey
Final Motor Drive	When given a target angle, motors must complete turn in correct direction	Using a test firmware, send a specific angle command to the robot. Motors will perform turn and visually verify the accuracy of the turn against a marked surface.	Untested	Lucas Kettler
Final Power Test	With the system under a full load, all regulated voltage outputs (5V, 12V) must remain stable and within **±5%** of their target values.	Connect an oscilloscope to microcontrollers, motors, and microphone array, and test motor actions with microcontroller acting. Measure voltage and current to ensure battery is powering the whole system.	Untested	Joshua Teele
Final Motor Control Signal Verification	When a navigation command is given the correct GPIO pins on the microcontroller must output the corresponding direction signal	Connect an oscilloscope to output pins of microcontroller, send commands, and verify the correct output	Untested	Sanjana Pasumarthi